High-Order Winslow smoothing/untangling

Andrew Shi

April 22, 2026

Contents

0 The 2016 formulation (brief recap) 2
0.1 Numerical Results/Tests 2

1 W modification to achieve reference element shape 3
1.1 Target metric 5
1.2 Modified split form 5
1.3 Continuity of W and the $\tilde{\alpha}$ projection 5
1.4 Numerical Results/Tests 6
1.5 Numerical experiments with the W-modification (negative result) 6

Codes live here.

0 The 2016 formulation (brief recap)

The Winslow equations in physical coordinates are

$$g^{ij}\,\partial_i\partial_j x_k = 0, \qquad k = 1, \ldots, n, \tag{1}$$

where g^{ij} are the contravariant metric components of the (nonlinear) mapping from the computational domain C to the physical domain D. The nonconservative form does not discretize cleanly in a CG framework, so the 2016 paper introduces auxiliary variables $\alpha_j = -\partial_i(g^{ij})$ and recasts the system in split form:

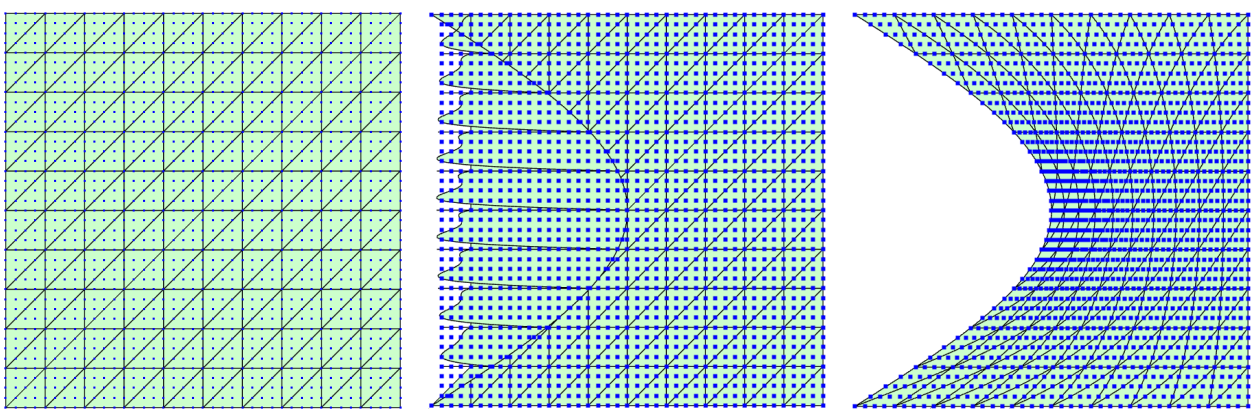
$$\partial_i(g^{ij}) + \alpha_j = 0, \tag{2}$$

$$\partial_i(g^{ij}\,\partial_j x_k) + \alpha_j\,\partial_j x_k = 0, \tag{3}$$

with Dirichlet data $\boldsymbol{x} = \boldsymbol{x}_{\text{bnd}}$ on ∂C. The point of this split form is to 1) mimic the what finite difference Winslow people are doing and 2) enable Picard iteration, which is the "magic" here.

0.1 Numerical Results/Tests

The code `w2.m` does the following in Figure 1.



(a) Reference mesh. (b) Initial configuration. (c) After Winslow

Figure 1: (a) Structured 11×11 reference mesh at $p = 4$. (b) Initial configuration: interior nodes unchanged, boundary on side 4 displaced by $0.5\sin(\pi y)$, producing heavy tangling. (c) After Winslow iterations, interior nodes have redistributed to absorb the boundary deformation; all elements valid.

To assess mesh quality we use the scaled Jacobian of Fortunato–Persson [1],

$$I \;=\; \frac{\min_{\xi\in\kappa} J(\xi)}{\max_{\xi\in\kappa} J(\xi)}, \qquad I \le 1, \tag{4}$$

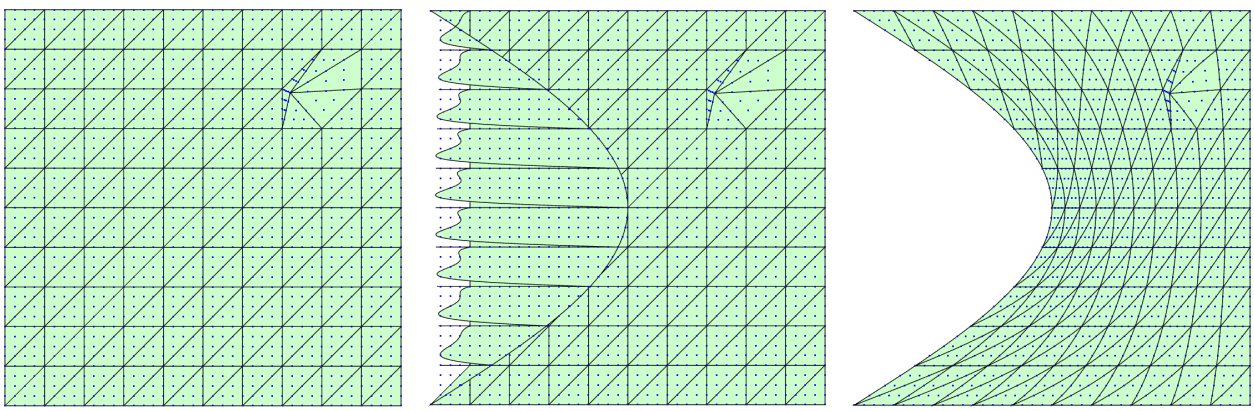
approximated at the Gauss quadrature points of each element κ. For a straight-sided simplex, J is constant across κ and $I = 1$ exactly; $I \le 0$ indicates a degenerate or inverted element.

To set up the main test case we will use throughout Section 1, we introduce `w2_skewed.m`, a slight variation of `w2.m` in which a single interior linear vertex of the otherwise uniform 11×11

reference is moved toward a neighbour, creating a small patch of sliver elements in the upper-right of the reference. Everything else—the boundary perturbation on side 4, the polynomial order, the smoother itself—is identical to `w2.m`. The scaled Jacobian I of (4) is blind to this regime: any simplex with straight edges has $I = 1$ regardless of aspect ratio, so it cannot "see" the sliver patch in the reference. To capture straight-sided shape distortion we also report the Liu–Joe inverse mean-ratio shape distortion metric [2, 3],

$$\eta \;=\; \frac{\|\boldsymbol{J}\|_F^2}{d\,|\det \boldsymbol{J}|^{2/d}}, \qquad \eta \in [1, \infty), \tag{5}$$

with $\eta = 1$ for a conformal (ideal) element and $\eta \to \infty$ as the element degenerates. The results are summarized in Figure 2 (meshes), Figure 3 (scaled Jacobian distribution), and Figure 4 (shape distortion distribution). Unmodified Winslow produces a valid physical mesh, but the sliver cluster visible in the reference persists as the worst element in the smoothed output. This is exactly the behaviour the W-modification of Section 1 is designed to address, and the single-sliver `w2_skewed` test is the setting in which we evaluate it.

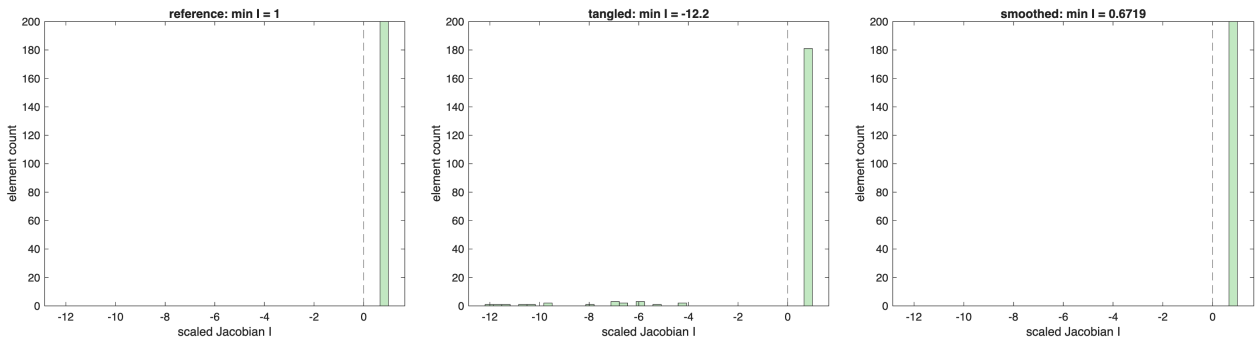


(a) Reference mesh. (b) Initial configuration. (c) After Winslow

Figure 2: Slight variation of `w2.m`: one interior vertex of the 11×11 reference at $p = 4$ is displaced toward a neighbour, creating a sliver cluster. (a) Skewed reference with $\max \eta = 4.63$. (b) Boundary on side 4 displaced by $0.5 \sin(\pi y)$, producing heavy tangling near the perturbed edge on top of the reference-induced sliver. (c) After unmodified Winslow: elements are valid, but the reference skew persists, with $\max \eta = 7.17$ on the sliver in the upper-right.

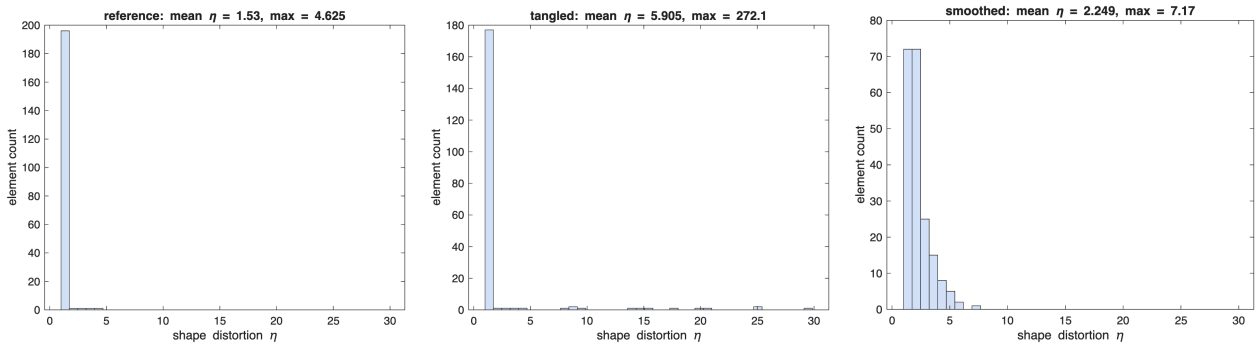
1 W modification to achieve reference element shape

The 2016 formulation produces a physical mesh whose element shapes are inherited from the reference mesh: the contravariant metric g^{ij} in the split-form system of Section 0 is built from the reference element geometry, so a skewed reference produces skewed physical elements. The W-modification below decouples the target shape from the reference element shape. **This is exactly the same formulation Persson gave me from his Gemini conversation (persson_gemini_winslow.pdf).**



(a) Reference (b) Tangled (c) Smoothed

Figure 3: Per-element scaled Jacobian $I = \min_\xi J(\xi)/\max_\xi J(\xi)$ for the `w2_skewed` test. (a) Reference: all elements straight-sided simplex, so $I = 1$ identically—the metric is blind to the sliver cluster present in the mesh. (b) Tangled: long negative tail, $\min I = -12.2$. (c) Smoothed: all $I > 0$, $\min I \approx 0.67$ on elements bordering the indentation.



(a) Reference (b) Tangled (c) Smoothed

Figure 4: Per-element shape distortion metric $\eta = \|J\|_F^2/(d\,|\det J|^{2/d})$ [2, 3] for the `w2_skewed` test. $\eta \in [1, \infty)$ with $\eta = 1$ at conformal (ideal) elements; unlike I, η detects straight-sided slivers. (a) Reference: $\max \eta = 4.63$, mean 1.53—the right tail is the sliver cluster around the moved vertex. (b) Tangled: $\max \eta = 272$. (c) Smoothed: $\max \eta = 7.17$, mean 2.25—the sliver cluster persists as the worst element in the smoothed output, motivating the W-modification in Section 1.

1.1 Target metric

For each reference element $K \in \mathcal{T}_h$, let $W_K \in \mathbb{R}^{n\times n}$ denote the Jacobian of an affine map from an ideal element (e.g. a unit equilateral simplex or unit square) to the reference element geometry. The total map from the ideal element to the physical element then has Jacobian $J_{\text{tot}} = JW_K^{-1}$, where $J = \partial x/\partial \xi$ is the reference-to-physical Jacobian.

The covariant and contravariant metrics of the total map are

$$\tilde{g}_{ij} = (J_{\text{tot}}^T J_{\text{tot}})_{ij} = (W_K^{-T} g W_K^{-1})_{ij}, \qquad \tilde{g}^{ij} = (W_K g^{-1} W_K^T)_{ij} = W_{K,ik}\, g^{kl}\, W_{K,jl}. \tag{6}$$

With $W_K = I$ for all K, $\tilde{g}^{ij} = g^{ij}$ and the 2016 formulation is recovered.

1.2 Modified split form

We replace g^{ij} by $\tilde{g}^{ij}$ in the split-form system of Section 0, introducing a modified auxiliary variable $\tilde{\alpha}_j = -\partial_i \tilde{g}^{ij}$:

$$\partial_i(\tilde{g}^{ij}) + \tilde{\alpha}_j = 0, \tag{7}$$

$$\partial_i(\tilde{g}^{ij}\, \partial_j x_k) + \tilde{\alpha}_j\, \partial_j x_k = 0, \tag{8}$$

with Dirichlet data $x = x_{\text{bnd}}$ on ∂C. The equivalence with the nonconservative form $\tilde{g}^{ij}\partial_i\partial_j x_k = 0$ follows from (7) by the same product-rule argument as in the unmodified case.

Heuristic nature of the extension. The classical Winslow system $g^{ij}\partial_i\partial_j x_k = 0$ is the reformulation, via change of variables, of $\Delta_x \xi_k = 0$—the Euler–Lagrange equation of the Dirichlet energy $\int_D |\nabla_x \xi|^2\, dx$ for the inverse map. The substitution $g \mapsto \tilde{g}$ in (7)–(8) is not, to the authors' knowledge, the Euler–Lagrange equation of any natural energy functional in x. In particular, for piecewise-constant W_K the map $\xi \mapsto W_K\xi$ is a per-element linear reparameterization of the reference coordinates, and harmonic maps are invariant under such reparameterizations, so the natural variational interpretation collapses to the unmodified Winslow system. We therefore regard (7)–(8) as a heuristic extension of the 2016 numerical scheme, parameterized so as to retain its Picard structure, rather than as the discretization of a derived PDE. Consequences: no Radó–Kneser–Choquet analog applies, and convergence of the resulting Picard iteration is assessed empirically.

1.3 Continuity of W and the $\tilde{\alpha}$ projection

The original Fortunato–Persson formulation already solves one projection problem: the auxiliary variable α_j is defined, through the weak form of $\partial_i g^{ij} + \alpha_j = 0$, as the L^2 projection of $-\partial_i g^{ij}$ onto the continuous CG space V_h^p. Operationally this is the linear solve $\alpha \leftarrow M\backslash b$, where M is the V_h^p mass matrix and b is assembled by quadrature from g^{ij} at Gauss points. The solve is what makes the scheme work on unstructured meshes. Because $x_h \in V_h^p$ is C^0 but not C^1, the gradient $\partial x/\partial\xi$ jumps across every internal edge, so g^{ij} is discontinuous there, and a strong derivative of g^{ij} would produce Dirac-style spikes supported on the interfaces. By omitting internal face fluxes in the weak form, the projection smears those spikes into a single continuous $\alpha \in V_h^p$, bypassing the DG machinery of numerical fluxes entirely. This is the computational magic of the 2016 paper.

In the modified system (7)–(8) this projection is reused without modification. $\tilde{\alpha}_j$ is the V_h^p projection of $-\partial_i \tilde{g}^{ij}$, computed with the *same* mass matrix M and the same backslash solve; only the right-hand side is altered to use $\tilde{g}^{ij} = W_K g^{ij} W_K^T$ instead of g^{ij}. No new projection infrastructure is introduced at this stage, and one can formally proceed without any further thought about the role of W_K.

The genuinely new object is W_K. It is the Jacobian of the affine map from the ideal equilateral triangle to the (straight-sided) reference triangle K—a map that sends the three corners of the ideal element to the three corners of K. Because the map is affine, its Jacobian W_K is a *single* 2×2 matrix, constant throughout the interior of K, at every polynomial order p; the polynomial order p controls how the *solution* x_h is represented, not the geometry of K, and the straight-sided reference keeps W_K element-constant regardless of p. The subscript K records the essential point that W_K is one matrix per element and jumps discontinuously across interfaces, since adjacent reference triangles are mapped from the ideal by different affine transformations.

At a glance it is tempting to conclude that these new jumps are absorbed for free by the $\tilde{\alpha}$-projection, just as the old jumps of g^{ij} were. The projection is indeed well-defined, but a subtler point intervenes. The jumps of g^{ij} at an interface are controlled by the jumps of $\partial x_h/\partial \xi$ there, and these shrink to zero as the Picard iteration converges, since the fixed point x_h is as smooth as the discrete space allows. FP's robustness relies on this self-limiting behaviour: the coefficient roughness that the iteration has to cope with decays alongside the iteration itself. The jumps of W_K are structurally different. They are set once and for all by the initial reference mesh, are external to the iterate, and *do not* decay with Picard. If the reference mesh contains neighbours of very different shape, the W_K jumps can be substantial, and we have observed empirically that the resulting persistent coefficient roughness destabilises the iteration.

The remedy suggested in the Persson/Gemini exchange is to elevate W to V_h^p *before* forming $\tilde{g}^{ij}$. At each CG node x_n, assemble a nodal target $W_h(x_n)$ as a weighted average of the W_K values from the surrounding elements,

$$W_h(\mathbf{x}_n) = \frac{\sum_{K \ni \mathbf{x}_n} |K|\, W_K}{\sum_{K \ni \mathbf{x}_n} |K|}, \tag{9}$$

then interpolate via the standard shape functions. The resulting $W_h \in V_h^p$ is C^0, so $\tilde{g}^{ij} = W_h g^{ij} W_h^T$ inherits *only* the jumps already present in g^{ij}, which the $\tilde{\alpha}$-projection handles as before. The averaging itself is cheap: it is a lumped-mass L^2 projection onto V_h^p, equivalent to area-weighted nodal averaging. A full (non-lumped) L^2 projection would require an additional global mass-matrix solve but is not necessary, since the Winslow-like operator is relatively insensitive to moderate blur in the target metric.

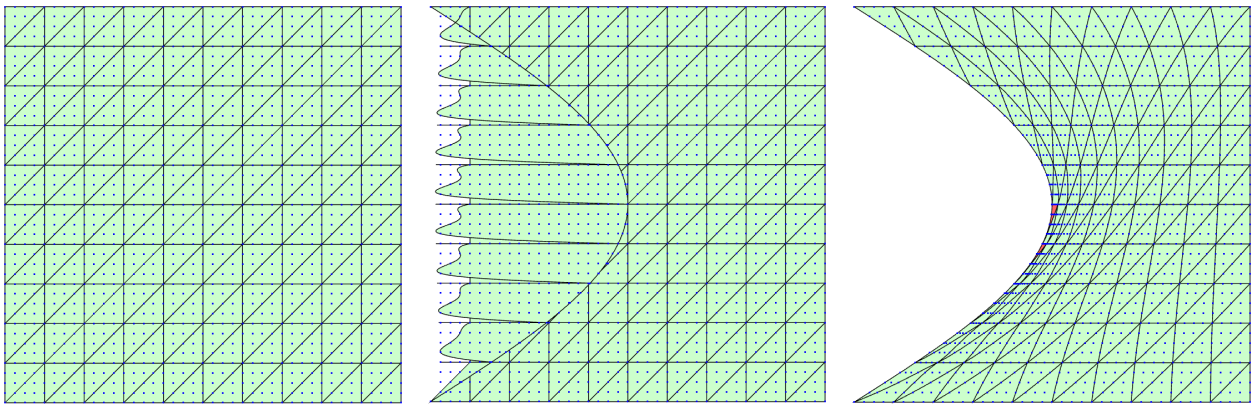
1.4 Numerical Results/Tests

We first run the unskewed `w2.m` setup of Section 0.1 through the W-modified solver (`w2_W.m`). The reference mesh, boundary perturbation, and polynomial order are identical to `w2.m`; only g^{ij} is replaced by $\tilde{g}^{ij} = W_h\, g^{ij}\, W_h^T$, with W_h obtained by full (consistent) L^2 projection of the equilateral-target W_K onto V_h^p—the variant closest to the original 2016 formulation. The result is in Figure 5, to be compared directly to Figure 1.

The smoothed mesh has $\min J = -4.4 \times 10^{-4}$: two interior elements are slightly inverted (highlighted in red in Figure 5(c)), where `w2.m` produced $\min J = +3.1 \times 10^{-3}$ and all elements valid. A regression test (`w_regression.m`) verifies that $W_h \equiv I$ reproduces the unmodified solver bit-for-bit, so the inversion is from the formulation, not the implementation.

1.5 Numerical experiments with the W-modification (negative result)

We implemented the split-form modification of Section 1—literally substituting $\tilde{g}^{ij} = W_K g^{-1} W_K^T$ in place of g^{ij} in both `sm1alpha2d` and `sm12d` of the Picard solver—and ran it on the single-sliver test of Figure 2. The implementation lives in `elliptic_smoothing_W.m`, with the per-



(a) Reference mesh. (b) Initial configuration. (c) After W-modified Winslow

Figure 5: `w2_W`: the `w2.m` setup run through the W-modified Picard solver with W_h constructed by consistent L^2 projection onto V_h^p. (a) and (b) are identical to their counterparts in Figure 1. In (c) the two red elements (centroids $(0.46, 0.37)$ and $(0.50, 0.47)$) have $\min J < 0$, so the W-modified output is inverted on a problem where the original 2016 formulation produced a valid mesh.

element W_K built by `compute_WK_simplex.m` from an ideal equilateral triangle. A regression test (`demos/w_regression.m`) verifies that the modified solver reproduces the unmodified output bit-for-bit when $W_K = I$ at every element, so the wiring is trustworthy.

The experimental conclusion is negative. The W-modification converges in most of the variants we tried, but none of them produce a smoothed mesh that is meaningfully better than the unmodified Winslow output on the sliver test. Below we report the specific variants we tried and why each fails.

Piecewise-constant W_K. The simplest implementation: feed the raw per-element W_K directly into the substituted split form, with no smoothing across element interfaces. For the uniform unskewed reference mesh this already diverges: adjacent triangles in each cell of `mshsquare` have different corner orderings, so W_K jumps by a rotation/reflection at every interior edge. The jumps in $\tilde{g}^{ij}$ that result are too large for the weak $\tilde{\alpha}$-projection to absorb, and the Picard iteration blows up. Restricting $W_K \neq I$ to only the four sliver-cluster elements, with $W_K = I$ everywhere else, also diverges—even four interfaces with a non-trivial jump are enough to destabilize.

Nodal projection onto P^1 (linear corners only). As suggested in Section 1.3, we project W_K to a continuous field W_h by a lumped L^2 projection onto the linear corner vertices of the mesh (Eq. (9)), then interpolate linearly to all DG nodes for use at Gauss points. This stabilizes the Picard iteration but also washes the correction out. On the single- sliver test, the sliver reference vertex at $\xi = (0.72, 0.79)$ is mapped by the smoothed output to: $(0.7964, 0.79)$ under unmodified Winslow and $(0.8029, 0.79)$ under W-modified (P^1-projected) Winslow. The target (fully un-skewed) is $(0.8, 0.8)$. The unmodified solver already performs 96% of the x-correction on its own by virtue of being a harmonic smoother; the W-modification contributes an additional 0.006 shift, within the noise floor of the iteration. Neither method moves the y-coordinate at all, because the boundary perturbation in `w2.m` is purely in x and the Winslow system decouples component-wise. Globally the W-modified output is in fact *worse* than the unmodified one: $\max \eta = 19.2$ versus 7.17, with the worst element having migrated from the sliver location to the lower-middle of the domain.

Nodal projection onto V_h^p (all CG nodes). The notes' derivation (Section 1.3) places $W_h \in V_h^p$, i.e., on the full order-p CG space, not just the linear corners. We implemented this variant as well. On the uniform unskewed mesh (`w2.m`) it converges but produces a smoothed mesh with $\min J = -2.6 \times 10^{-4}$, i.e. *inverted* elements, on a problem where the unmodified solver has no difficulty. On the sliver test the iteration fails to converge altogether. Full V_h^p preserves more of the per-element W_K near element interiors and less of the corner averaging, and this extra expressiveness is enough to push the heuristic past stability.

Weighting schemes in the lumped projection. Equation (9) uses the element area $|K|$ as the lumping weight. This down-weights slivers (which have small $|K|$) relative to their regular-sized neighbors, so the sliver's W_K is largely drowned out in the projected W_h. We replaced $|K|$ by η_K, $\eta_K - 1$, and η_K^2 in an attempt to boost the sliver's contribution, but the sliver vertex image remained within ± 0.01 of the unmodified result across all schemes, and global $\max \eta$ stayed between 19 and 22. The weighting choice does not meaningfully move the numerical outcome.

Sliver-only W. Restricting $W_K \neq I$ to only the four sliver-cluster elements (identified by an η threshold on the reference) and projecting only those via P^1 gives a sliver image of $(0.7966, 0.79)$—indistinguishable from the unmodified $(0.7964, 0.79)$. The averaging with the identity-valued W_K of the surrounding elements at shared CG corners neutralizes the sliver correction almost entirely. The piecewise-constant and V_h^p variants of this experiment both diverge.

Summary. Table 1 collects the results. Across every variant we tried—three projection choices (piecewise constant, P^1, V_h^p), four lumping weights (area, η, $\eta - 1$, η^2), and a sliver-only mask—the W-modification either diverges, or converges to an output whose sliver image agrees with the unmodified Winslow result to within the iteration's noise floor. The reason is visible by inspection: the unmodified Winslow solver, being harmonic, already pulls the interior sliver vertex 96% of the way back to the un-skewed target. Any projection that is stable enough to keep Picard converging is also one that averages the local W_K correction heavily with its identity-valued neighbors, leaving only a few-tenth-of-a-percent shift that cannot close the remaining 4% gap.

Table 1: Sliver reference vertex $\xi = (0.72, 0.79)$ image under different W-modification variants, on the single-sliver test of Figure 2. "Target" denotes the un-skewed position $(0.8, 0.8)$. Unmodified Winslow is shown for reference.

variant	sliver image	global $\max \eta$	notes
unmodified Winslow	$(0.7964, 0.79)$	7.17	baseline
piecewise-const W_K	—	—	diverges
P^1 projection, area weight	$(0.8029, 0.79)$	19.2	converges
P^1 projection, η weight	$(0.8010, 0.79)$	19.5	converges
P^1 projection, η^2 weight	$(0.7937, 0.79)$	21.7	converges
P^1 projection, $\eta - 1$ weight	—	—	diverges
V_h^p projection	—	—	diverges or $\min J < 0$
sliver-only W, P^1	$(0.7966, 0.79)$	7.05	$\approx$ unmodified
sliver-only W, piecewise const	—	—	diverges
sliver-only W, V_h^p	—	—	diverges
target (un-skewed)	$(0.80, 0.80)$	—	

Interpretation. This is consistent with the caveat already stated in Section 1.2: the modified split form is not the Euler–Lagrange equation of any natural energy, and there is no discrete minimum principle guaranteeing that the Picard fixed point is a meaningful "W-harmonic" map. Empirically, for interior sliver defects on an otherwise well-shaped mesh, the W-modification as written in Section 1 does not deliver the reference-agnostic behaviour that the classical formulation lacks—it just reshuffles the distortion without reducing it. Whether the construction can be rescued by machinery outside the scope of Section 1 (for instance an anisotropic spatially-varying ideal, damped/continuation Picard, or a proper variational reformulation with an H^1-type energy in x) is an open question.

References

[1] Meire Fortunato and Per-Olof Persson. High-order unstructured curved mesh generation using the Winslow equations. *Journal of Computational Physics*, 307:1–14, 2016.

[2] Anwei Liu and Barry Joe. Relationship between tetrahedron shape measures. *BIT Numerical Mathematics*, 34(2):268–287, 1994.

[3] A. Shi and P.-O. Persson. Local element operations for curved simplex meshes. *International Journal for Numerical Methods in Engineering*, 2023. Author manuscript.